Constructing lenses in double categories

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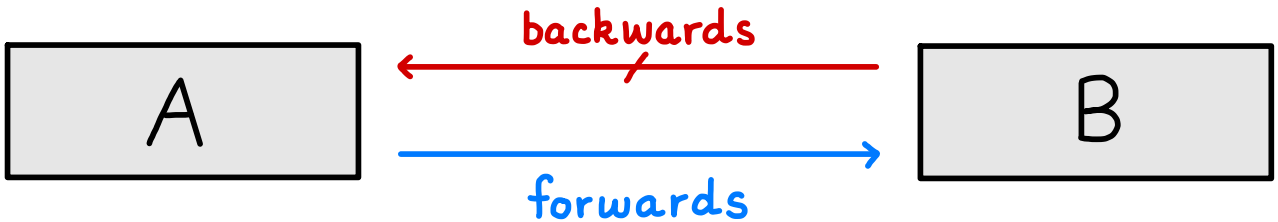
MOTIVATION

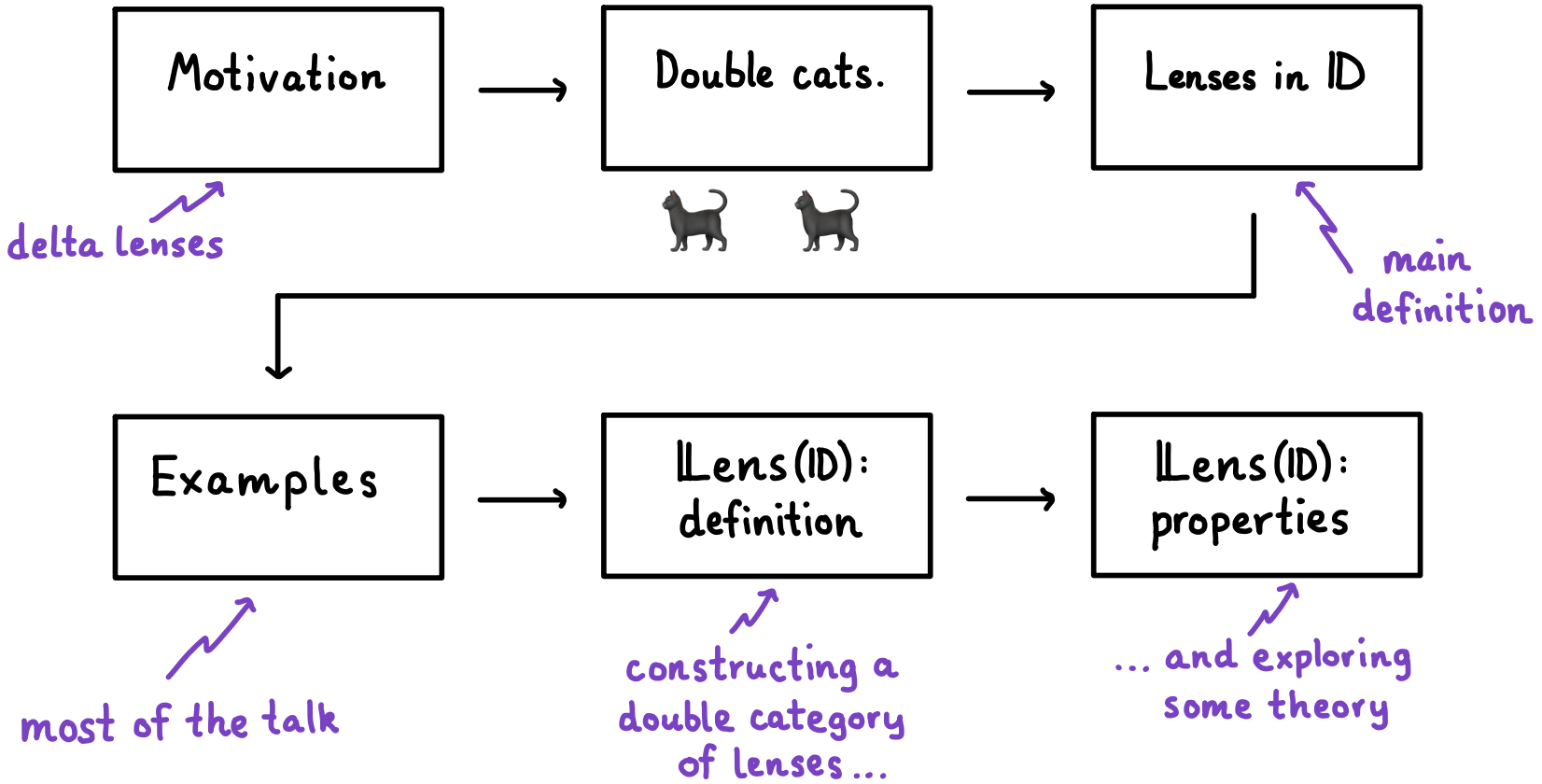
- A lens is a morphism consisting of two components:



- What if the forwards and backwards components of a lens were the horizontal and vertical morphisms in a double category?

$$\begin{tikzcd}
A \arrow[r, equal] \arrow[d, "/"] \arrow[dr, phantom, "\propto"] & A \arrow[d, equal] \\
B \arrow[r] & A
\end{tikzcd}$$

OVERVIEW OF THE TALK



MOTIVATING EXAMPLE: FUNCTORS & COFUNCTORS

A cofunctor $(f,\varphi): A \longrightarrow\!\!\!

WHAT IS A DOUBLE CATEGORY?

A double category $\mathbb{D}$ consists of:

- objects A, B, C, D, etc
- horizontal morphisms $\bullet \longrightarrow \bullet$
- vertical morphisms $\bullet \longrightarrow\!\!\!

DOUBLE CATEGORY OF FUNCTORS & COFUNCTORS

A square of functors and cofunctors,

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "{(f,\varphi)}"'] \arrow[dr, phantom, "\swarrow"] & C \arrow[d, "{(g,\gamma)}"] \arrow[dl, phantom, "\swarrow"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

is compatible if:

(1) For all $a \in A$, $gha = kfa$

(2) For all $(a \in A, u : fa \to b \in B)$,

$$h\varphi(a,u) = \gamma(ha, ku)$$

<u>Note</u>: uniquely determined by boundary of square.

Let $\mathbb{C}\mathrm{of}$ be the double cat. whose:

- objects are categories
- horizontal morphisms are functors
- vertical morphisms are cofunctors
- cells are compatible squares

A delta lens is a cell in $\mathbb{C}\mathrm{of}$:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "{(f,\varphi)}"'] \arrow[dr, phantom, "\swarrow"] & B \arrow[d, "1_B"] \arrow[dl, phantom, "\swarrow"] \\ B \arrow[r, equal] & B \end{tikzcd}$$

LENSES IN DOUBLE CATEGORIES

A lens $(f, \alpha, f') : A \nrightarrow B$ in a double category $\mathbb{D}$ is a cell of the form:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f'"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "1_B", "\shortmid"{marking}] \\ B \arrow[r, equal] & B \end{tikzcd}$$

If $\mathbb{D}$ is flat, then lenses in $\mathbb{D}$ express compatibility between horizontal and vertical morphisms.

Lenses in $\mathbb{D}^v$

Cells in $\mathbb{D}$ $\rightsquigarrow$

$$\begin{tikzcd} B \arrow[r, equal] \arrow[d, "f'"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "1_B", "\shortmid"{marking}] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

Lenses in $\mathbb{D}^h$

Cells in $\mathbb{D}$ $\rightsquigarrow$

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "1_B"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "f'", "\shortmid"{marking}] \\ A \arrow[r, equal] & A \end{tikzcd}$$

Lenses in $\mathbb{D}^{vh}$

Cells in $\mathbb{D}$ $\rightsquigarrow$

$$\begin{tikzcd} A \arrow[r, equal] \arrow[d, "1_A"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & A \arrow[d, "f'", "\shortmid"{marking}] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

SPLIT EPIMORPHISMS AS LENSES

Let $\mathcal{C}$ be a category and $\mathbb{S}q(\mathcal{C})$ be the double category whose:

- objects are those of $\mathcal{C}$.
- horizontal and vertical morphisms are those in $\mathcal{C}$.
- cells are commutative squares in $\mathcal{C}$:

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] \arrow[dr, phantom, "\circlearrowright"] & C \arrow[d, "g"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

Lenses in $\mathbb{S}q(\mathcal{C})$ are boring; they are just morphisms in $\mathcal{C}$:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] \arrow[dr, phantom, "\circlearrowright"] & B \arrow[d, "1_B"] \\ B \arrow[r, "1_B"'] & B \end{tikzcd}$$

Lenses in $\mathbb{S}q(\mathcal{C})^{\vee}$ are split epimorphisms:

$$\begin{tikzcd} A \arrow[r, "1_A"] \arrow[d, "\sigma"'] \arrow[dr, phantom, "\circlearrowright"] & A \arrow[d, "1_A"] \\ B \arrow[r, "\rho"'] & A \end{tikzcd}$$

ISOMORPHISMS & MONOMORPHISMS

Let $\mathcal{C}$ be a category and $\mathbb{P}\mathrm{b}(\mathcal{C})$ be the double category whose:

- objects are those of $\mathcal{C}$.
- horizontal and vertical morphisms are those in $\mathcal{C}$.
- cells are pullback squares in $\mathcal{C}$:

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] \arrow[dr, phantom, "\lrcorner", very near start] & C \arrow[d, "g"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

Lenses in $\mathbb{P}\mathrm{b}(\mathcal{C})$ are isomorphisms in $\mathcal{C}$:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "1_B"] \\ B \arrow[r, "1_B"'] & B \end{tikzcd}$$

Lenses in $\mathbb{P}\mathrm{b}(\mathcal{C})^{vh}$ are monomorphisms:

$$\begin{tikzcd} A \arrow[r, "1_A"] \arrow[d, "1_A"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

LENSES IN 2-CATEGORIES

Let $\mathcal{K}$ be a 2-category and let $\mathbb{Q}(\mathcal{K})$ be the double category whose:

- objects are those of $\mathcal{K}$.
- horizontal and vertical morphisms are those in $\mathcal{K}$.
- cells are given by diagrams in $\mathcal{K}$:

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & C \arrow[d, "g"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

Lenses in $\mathbb{Q}(\mathcal{K})$ are natural transformations:

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & B \arrow[d, "1_B"] \\ B \arrow[r, "1_B"'] & B \end{tikzcd}$$

Lenses in $\mathbb{Q}(\mathcal{K})^v$ and $\mathbb{Q}(\mathcal{K})^h$ are given by:

$$\begin{tikzcd} A \arrow[r, "1_A"] \arrow[d, "\sigma"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & A \arrow[d, "1_A"] \\ B \arrow[r, "\rho"'] & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "\sigma"] \arrow[d, "1_A"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & C \arrow[d, "\rho"] \\ A \arrow[r, "1_A"'] & A \end{tikzcd}$$

LENSES IN BICATEGORIES

Let $\mathcal{K}$ be a bicategory and let $\mathbb{V}(\mathcal{K})$ be the double category whose:

- objects are those of $\mathcal{K}$.
- horizontal morphisms are identities
- vertical morphisms are those in $\mathcal{K}$.
- cells are given by 2-cells in $\mathcal{K}$:

$$\begin{tikzcd} A \arrow[r, equal] \arrow[d, "f"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & C \arrow[d, "g"] \\ B \arrow[r, equal] & D \end{tikzcd}$$

Lenses in $\mathbb{V}(\mathcal{K})$ are copointed endofunctors:

$$\begin{tikzcd} A \arrow[r, equal] \arrow[d, "f"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & A \arrow[d, "1_A"] \\ A \arrow[r, equal] & A \end{tikzcd}$$

Lenses in $\mathbb{V}(\mathcal{K})^h$ are pointed endofunctors:

$$\begin{tikzcd} A \arrow[r, equal] \arrow[d, "1_A"'] \arrow[dr, phantom, "\overset{\alpha}{\Rightarrow}"] & A \arrow[d, "f"] \\ A \arrow[r, equal] & A \end{tikzcd}$$

SPLIT MULTI-VALUED FUNCTIONS

Let $\mathbb{S}\mathrm{pan}$ be the double category whose:

- objects are sets
- horizontal morphisms are functions
- vertical morphisms are spans
- cells are diagrams:

$$\begin{tikzcd}
A \arrow[r, "h"] & C \\
X \arrow[u, "f_1"] \arrow[r, "j"] \arrow[d, "f_2"'] & Y \arrow[u, "g_1"'] \arrow[d, "g_2"] \\
B \arrow[r, "k"'] & D
\end{tikzcd}$$

Lenses in $\mathbb{S}\mathrm{pan}^{vh}$ are split multi-valued functions:

$$\begin{tikzcd}
C \arrow[r, equal] & C \\
C \arrow[u, "1_C"] \arrow[r, "\sigma"] \arrow[d, "1_C"'] & Y \arrow[u, "g_1"'] \arrow[d, "g_2"] \\
C \arrow[r, "g_2\sigma"'] & D
\end{tikzcd}$$

<u>Fun fact</u>: The double category of split multi-valued functions classifies delta lenses! Delta lenses into $B \simeq$ lax double functors $\mathbb{V}(B) \longrightarrow \mathbb{S}\mathrm{Mult}$.

LENSES FROM COMONADS

Let $M : \mathcal{C} \to \mathcal{C}$ be a comonad and let $\mathbb{C}\mathrm{oKl}(M)$ be the double category whose:

- objects are those of $\mathcal{C}$
- horizontal morphisms are those of $\mathcal{C}$
- vertical morphisms are coKleisli maps
- cells are commutative squares in $\mathcal{C}$:

$$\begin{tikzcd} MA \arrow[r, "Mh"] \arrow[d, "f"'] & MC \arrow[d, "g"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

A lens in $\mathbb{C}\mathrm{oKl}(M)^{\vee}$ is a morphism with an M-section:

$$\begin{tikzcd} MA \arrow[r, equal, "M(1_A)"] \arrow[d, "\sigma"'] & MA \arrow[d, "\varepsilon_A"] \\ B \arrow[r, "p"'] & A \end{tikzcd}$$

For the comonad $M \times (-) : \mathrm{Set} \to \mathrm{Set}$ we have:

$$\begin{tikzcd} M \times B \arrow[r, equal] \arrow[d, "p"'] & M \times B \arrow[d, "\pi_A"] \\ A \arrow[r, "g"'] & B \end{tikzcd}$$

CONSTRUCT YOUR OWN EXAMPLE

There are many possible examples that one can construct. Two options:

1. Pick your favourite double category $\mathbb{D}$ and compute lenses in it.

- $\mathbb{P}\mathrm{rof}$
- $\mathbb{B}\mathrm{iLens}(\mathcal{C},\otimes,I)$
- $\mathbb{P}\mathrm{oly}$
- $\mathbb{R}\mathrm{el}$
- $\mathbb{O}\mathrm{ptic}(\mathcal{C},\otimes,I)$
- $\mathbb{M}\mathrm{nd}_{\mathrm{ret}}(\mathbb{D})$
- $\mathbb{T}\mathrm{op}$
- $\mathbb{A}\mathrm{rena}$
- Free double cat on $H \longleftarrow C_0 \longrightarrow V$

2. Pick your favourite lens and try to construct it from a double category.

<u>Limitation</u>: The backwards components of your lens must be composable independently of the lens composition!

<u>Non-example</u>: Composition of bimorphic lenses $(A,A') \xrightarrow{(f,p)} (B,B') \xrightarrow{(g,q)} (C,C')$.

$$A\times C' \xrightarrow{\Delta\times 1} A\times A\times C' \xrightarrow{1\times f\times 1} A\times B\times C' \xrightarrow{1\times q} A\times B' \xrightarrow{p} A'$$

THE DOUBLE CATEGORY OF LENSES

Let $\mathbb{L}\mathrm{ens}(\mathbb{D})$ be the double category whose:

- objects are those of $\mathbb{D}$
- horizontal morphisms are those of $\mathbb{D}$
- vertical morphisms are lenses
- cells with boundary

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "{(f,\alpha,f')}"'] \arrow[dr, phantom, "\theta"] & C \arrow[d, "{(g,\beta,g')}"] \\
B \arrow[r, "k"'] & D
\end{tikzcd}$$

are cells θ in $\mathbb{D}$ such that:

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "f'"'] \arrow[dr, phantom, "\theta"] & C \arrow[r, "g"] \arrow[d, "1"] \arrow[dr, phantom, "\beta"] & D \arrow[d, "1_D"] \\
B \arrow[r, "k"'] & D \arrow[r, equal] & D
\end{tikzcd}
=
\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "f'"'] \arrow[dr, phantom, "\alpha"] & B \arrow[r, "k"] \arrow[d, "1"] \arrow[dr, phantom, "1_k"] & D \arrow[d, "1_D"] \\
B \arrow[r, equal] & B \arrow[r, "k"'] & D
\end{tikzcd}$$

Composition of lenses is given by:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "f'"'] \arrow[dr, phantom, "\alpha"] & B \arrow[r, "g"] \arrow[d, "1_B"] \arrow[dr, phantom, "1_g"] & C \arrow[d, "1_C"] \\
B \arrow[r, equal] \arrow[d, "g'"'] \arrow[dr, phantom, "1_{g'}"] & B \arrow[r, "g"] \arrow[d, "g'"'] \arrow[dr, phantom, "\beta"] & C \arrow[d, "1_C"] \\
C \arrow[r, equal] & C \arrow[r, equal] & C
\end{tikzcd}$$

UNIVERSAL PROPERTY OF LENSES

A double category $\mathbb{D}$ is right-connected if for every vertical morphism $f : A \to\!\!\!

CANONICAL DOUBLE FUNCTORS

There are forgetful double functors:

$$\mathbb{D} \xleftarrow[\text{backwards component}]{\mathcal{L}} \mathbb{L}\mathrm{ens}(\mathbb{D}) \xrightarrow[\text{forwards component}]{\mathcal{U}} \mathbb{S}\mathrm{q}(\mathbb{D}_0)$$

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f'"', "|" marking] \arrow[dr, phantom, "\theta"] & C \arrow[d, "g'", "|" marking] \\ B \arrow[r, "k"'] & D \end{tikzcd} \rightsquigarrow \begin{tikzcd} A \arrow[r, "h"] \arrow[d, "(f,\alpha,f')"', "|" marking] \arrow[dr, phantom, "\theta"] & C \arrow[d, "(g,\beta,g')", "|" marking] \\ B \arrow[r, "k"'] & D \end{tikzcd} \rightsquigarrow \begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] \arrow[dr, phantom, "\circlearrowright"] & C \arrow[d, "g"] \\ B \arrow[r, "k"'] & D \end{tikzcd}$$

What if a lens was a backwards component with coalgebraic structure, or a forwards component with algebraic structure?

COALGEBRAIC PERSPECTIVE

A double category $\mathbb{D}$ is equivalent to an internal category in $\mathcal{C}at$:

$$\begin{tikzcd} \mathcal{D}_1 \times_{\mathcal{D}_0} \mathcal{D}_1 \arrow[r, "\mathrm{comp}"] & \mathcal{D}_1 \arrow[r, shift left, "\mathrm{dom}"] \arrow[r, shift right, "\mathrm{cod}"'] & \mathcal{D}_0 \arrow[l, "\mathrm{id}"'] \end{tikzcd}$$

where:

$\mathcal{D}_0$ - cat. of objects & hori. morph.

$\mathcal{D}_1$ - cat. of vert. morph. & cells

<u>Assumptions</u>:

(1) $\mathrm{cod}: \mathcal{D}_1 \longrightarrow \mathcal{D}_0$ has a RAR1

(2) Each fibre has finite products

<u>Conjecture</u>: The functor

$$\mathcal{L}ens(\mathbb{D}) \xrightarrow{\mathcal{L}_1} \mathcal{D}_1$$

is comonadic.

<u>Idea</u>: Given $f: A \nrightarrow B$, take product in fibre over B with $1_B: B \nrightarrow B$:

$$\begin{tikzcd} A \times_B B \arrow[r, "\pi_B"] \arrow[d, "f \times_B 1_B"'] \arrow[dr, phantom, "\pi_{1_B}"] & B \arrow[d, "1_B"] \\ B \arrow[r, equal] & B \end{tikzcd}$$

cofree lens

• Conjecture is true for $\mathbb{D} = \mathbb{C}of$.

ALGEBRAIC PERSPECTIVE

Every right-connected double category $\mathbb{C}$ admits a canonical double functor:

$$\mathbb{C} \xrightarrow{u} \mathbb{S}q(\mathcal{C}_0)$$

Bourke & Garner showed that

$$\mathcal{C}_1 \xrightarrow{u_1} \mathcal{S}q(\mathcal{C}_0) = \mathcal{C}_0^{\mathbb{2}}$$

is monadic if and only if $\mathcal{C}_0$ admits an algebraic weak factorisation system.

<u>Assumptions</u>:

(1) $\mathrm{dom} : \mathbb{D}_1 \longrightarrow \mathbb{D}_0$ has a LARI

(2) $\mathrm{cod} : \mathbb{D}_1 \longrightarrow \mathbb{D}_0$ is an opfibration

<u>Conjecture</u>: The functor

$$\mathcal{L}ens(\mathbb{D}) \xrightarrow{u_1} \mathcal{S}q(\mathbb{D}_0)$$

has a left adjoint.

- Conjecture is true for $\mathbb{D} = \mathbb{C}of$.
- What about monadicity?

A COLLECTION OF PROPERTIES

The double category $\mathbb{L}\mathrm{ens}(\mathbb{D})$ has many nice properties:

If $\mathbb{D}$ is/has $\boxed{}$ then $\mathbb{L}\mathrm{ens}(\mathbb{D})$ is/has $\boxed{}$:

- Flat
- Companions
- Horizontal invariant
- Unit-pure with tabulators

- Conjoints in $\mathbb{L}\mathrm{ens}(\mathbb{D})$ arise precisely from horizontal isos.

- If $\mathbb{D} = \mathbb{C}\mathrm{of}$, then the codomain map $\mathrm{cod} \cdot \mathcal{L}\mathrm{ens}(\mathbb{C}\mathrm{of}) \longrightarrow \mathcal{C}\mathrm{at}$ is a bifibration $\Rightarrow$ pullback and pushforward of delta lenses along functors exist.

SUMMARY & FUTURE WORK

- We defined lenses in a double category via certain cells:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f'"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "1_B", "\shortmid"{marking}'] \\ B \arrow[r, equal] & B \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, equal] \arrow[d, "f'"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "1_B", "\shortmid"{marking}'] \\ A \arrow[r, "f"'] & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "f"] \arrow[d, "1_B"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & B \arrow[d, "f'", "\shortmid"{marking}'] \\ A \arrow[r, equal] & A \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, equal] \arrow[d, "1_A"', "\shortmid"{marking}] \arrow[dr, phantom, "\alpha"] & A \arrow[d, "f'", "\shortmid"{marking}'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

- We considered several examples, including delta lenses.

- We constructed a double category $\mathbb{L}\mathrm{ens}(\mathbb{D})$ and studied its properties as the right-connected completion.

- <u>Future work</u>: More examples, extend framework, and prove (co)monadicity conjectures.